

Database Foundations

3-4: Data Modeling Terminology and Mapping Practices

Exercise 1: Identify entities, attributes, instances and their corresponding tables, rows and columns

Overview

In this practice you identify entities, attributes, instances and their corresponding tables, rows and columns from the given scenarios.

Tasks

- Match the ERD elements to their corresponding database elements.

Analysis	Design
1. Attribute	a. Column
2. Entity	b. Foreign key
3. ER Model	c. Physical design
4. Instance	d. Primary key
5. Primary UID	e. Row
6. Relationship	f. Table
7. Secondary UID	g. Unique key

- Identify the table diagram notations listed below.
 - pk
 - fk
 - uk
 - *
 - o
- Create short names for the terms below based on the naming conventions rules.
 - Authors
 - Publishers
 - Customers
- The goal of this practice is to recognize attributes for an entity.

These three entities—SONG, EVENT, and CUSTOMER—play a role in a DJ business and are listed as the first three column headings in the table below. The fourth column contains a list of attributes. Use an X or a check mark to indicate that the attribute could belong to one or more of the entities listed. For example, could Title be an attribute for Song, for Event, and/or for Customer?

SONG	EVENT	CUSTOMER	
			Title
			Description
			Venue
			First Name
			Phone Number
			Release date
			Last Name
			Type
			Email address

Exercise 2: Mapping the Academic Database

Overview

In this practice you identify entities, attributes and their corresponding tables and columns from the given scenario.

Tasks

1. With the ERD provided below, map the entities, attributes and UIDs to tables, rows, and keys using a table diagram as shown:

